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1  CREATE TABLE suppliers(
2  supplier_id INT NOT NULL AUTO_INCREMENT,
3  name VARCHAR(50) NOT NULL,
4  phone VARCHAR(20) NOT NULL,
5  fax VARCHAR(20) NULL DEFAULT NULL,
6  nif CHAR(9) UNIQUE NOT NULL,
7  PRIMARY KEY(supplier_id)
8  );
9  -- one-to-many
10 CREATE TABLE brands(
11 brand_id INT NOT NULL AUTO_INCREMENT,
12 supplier_id INT NOT NULL,
13 name VARCHAR(50) UNIQUE,
14 PRIMARY KEY(brand_id),
15 CONSTRAINT fk_brands_supplier FOREIGN KEY(supplier_id) REFERENCES suppliers(supplier_id)
16 );
17 -- one-to-many
18 CREATE TABLE glasses(
19 glass_id INT NOT NULL AUTO_INCREMENT,
20 brand_id INT NOT NULL,
21 model VARCHAR(50) NOT NULL,
22 prescription_left DECIMAL(4,2) NOT NULL,
23 prescription_right DECIMAL(4,2) NOT NULL,
24 frame_type ENUM('Frameless', 'Acetate', 'Metallic'),
25 frame_color VARCHAR(20) NOT NULL,
26 left_lens_color VARCHAR(20) NOT NULL,
27 right_lens_color VARCHAR(20) NOT NULL,
28 cost_unit_price DECIMAL(8,4) NOT NULL,
29 stock TINYINT NOT NULL,
30 PRIMARY KEY(glass_id),
31 CONSTRAINT fk_glasses_brands FOREIGN KEY(brand_id) REFERENCES brands(brand_id)
32 );
33 -- one-to-one
34 CREATE TABLE addresses(
35 address_id INT NOT NULL AUTO_INCREMENT,
36 supplier_id INT UNIQUE NOT NULL,
37 street VARCHAR(50) NOT NULL,
38 address_number VARCHAR(10) NOT NULL,
39 address_floor VARCHAR(10) NULL,
40 door VARCHAR(10) NULL,
41 city VARCHAR(50) NOT NULL,
42 postal_code VARCHAR(10) NOT NULL,
43 country CHAR(2) NOT NULL,
44 PRIMARY KEY(address_id),
45 CONSTRAINT fk_addresses_suppliers FOREIGN KEY(supplier_id) REFERENCES suppliers(
46 supplier_id)
47 );
48 -- self-referencing
49 CREATE TABLE customers(
50 customer_id INT NOT NULL AUTO_INCREMENT,
51 recommended_by_id INT NULL,
52 is_active BOOLEAN NOT NULL,
53 name VARCHAR(50) NOT NULL,
54 postal_code VARCHAR(10) NOT NULL,
55 phone VARCHAR(20) NOT NULL,
56 email VARCHAR(100) UNIQUE NOT NULL,
57 registered_at DATE NOT NULL,
58 PRIMARY KEY(customer_id),
59 CONSTRAINT fk_customers_customers FOREIGN KEY(recommended_by_id) REFERENCES customers(
60 customer_id)
61 );
62 CREATE TABLE employees(
63 employee_id INT NOT NULL AUTO_INCREMENT,
64 first_name VARCHAR(50) NOT NULL,
65 last_name VARCHAR(50) NOT NULL,
66 PRIMARY KEY(employee_id)
67 );
68 -- many-to-many
69 CREATE TABLE orders(

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70 order_id INT NOT NULL AUTO_INCREMENT,
71 employee_id INT NOT NULL,
72 customer_id INT NOT NULL,
73 ordered_at DATETIME NOT NULL,
74 PRIMARY KEY(order_id),
75 CONSTRAINT fk_orders_employees FOREIGN KEY(employee_id) REFERENCES employees(employee_id),
76 CONSTRAINT fk_orders_customers FOREIGN KEY(customer_id) REFERENCES customers(customer_id)
77 );
78 -- many-to-many
79 CREATE TABLE details(
80 order_id INT NOT NULL,
81 glass_id INT NOT NULL,
82 quantity INT NOT NULL,
83 sale_unit_price DECIMAL(8,4) NOT NULL,
84 PRIMARY KEY(order_id, glass_id),
85 CONSTRAINT fk_details_orders FOREIGN KEY(order_id) REFERENCES orders(order_id),
86 CONSTRAINT fk_details_glasses FOREIGN KEY(glass_id) REFERENCES glasses(glass_id)
87 );
```